Grambank data offers some support for the view that sociopolitical complexity impacts on grammatical complexity

Sihan Chen*[1], David Gil[2], and Antonio Benítez-Burraco[3]

*Corresponding Author: sihanc@mit.edu
[1]Department of Brain and Cognitive Sciences, MIT, Cambridge, MA, USA
[2]Independent scholar
[3]Department of Spanish, Linguistics, and Theory of Literature (Linguistics), FAculty of Philology, University of Seville, Seville, Spain

Languages exhibit different degrees of complexity in their fundamental components (Sampson et al., 2009; McWhorter, 2011), with factors external to language, either physical or sociopolitical, possibly promoting or reducing complexity in specific language domains (Lupyan & Dale, 2016; Sinnemäki, 2020). Focusing on the latter, it has been proposed that close-knit or esoteric human groups speak languages with simpler syntax, and reduced compositionality and semantic transparency, but more complex morphology with more irregularity, whereas the languages of human groups forming extensive but loose social networks and involved in frequent contacts and cultural exchanges with other groups would show the opposite features (Bolender, 2007; Wray & Grace, 2007). Evidence for this claim is of a mixed and somewhat conflicting nature. Shcherbakova et al. (2023) found that languages spoken by exoteric societies tend to exhibit increased morphological fusion and informativity. In contrast, using the whole set of WALS (Dryer & Haspelmath, 2013) grammatical data, Chen et al. (2024) suggested that the languages spoken by esoteric societies tend towards less complex syntax, while the languages spoken by exoteric societies tend towards less complex morphology. In this paper, we replicate Chen et al.'s approach (2024) using data from Grambank (Skirgård et al., 2023), the largest comparative grammatical database available.

In our analysis, we examined the relation between morphological and syntactic complexity, and the sociopolitical complexity of the environment in which the language is spoken. Sociopolitical features were drawn from D-Place (Kirby et al., 2016), Glottolog (Hammarström et al., 2022), and Ethnologue

(Eberhard et al. (2022). A principal component analysis on these data yields a sociopolitical complexity index (SCI). **The more negative the SCI, the more complex the society.** The grammatical features were from Grambank (Skirgård et al., 2023), mostly presented as questions (e.g. Are there definite or specific articles?), and they are either present or absent in a language. We classified each feature as morphological and/or syntactic, and whether its presence increases or decreases complexity. Of the 195 features on Grambank, 99 are morphological, 66 syntactic, and 30 neither. We then conducted a mixed-effects logistic regression using the brms package (Bürkner, 2017) in R (R core team, 2021). To address Galton's problem (Roberts and Winters, 2013), we controlled for language relatedness and geographical proximity, following the approach in Shcherbakova et al. (2023). If the Chen et al. (2024) findings generalize from WALS to Grambank, we would expect **a positive correlation between SCI and morphological complexity, and a negative one between SCI and syntactic complexity**. We extracted all estimated effects between SCI and individual morphological and syntactic features (Figure 1) and calculated their averages. On average, sociopolitical complexity has a negative effect on morphological complexity and a small but positive effect on syntactic complexity, both robust after controlling for Galton's problem. Although limited in magnitude, our results thus offer some support for the predictions made in Chen et al. (2024).

The correlations between sociopolitical and grammatical complexity found here bear substantial phylogenetic implications. Since, until recently, all human societies were of lower levels of sociopolitical complexity, it may be inferred that their languages were also endowed with the properties associated with such societies, namely, higher morphological but lower syntactic complexity.

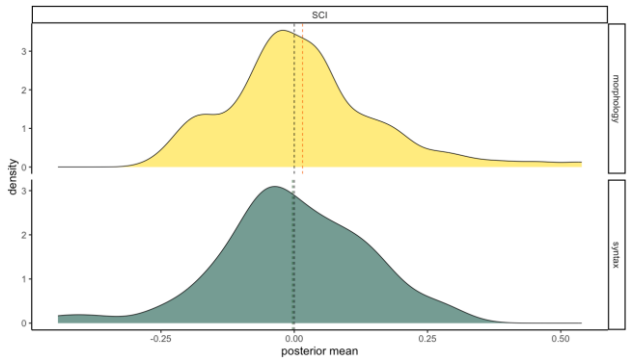

Figure 1. Distribution of posterior estimates for the effect of the sociopolitical index (SCI) on grammatical features. Negative posterior means indicate features that increase with sociopolitical complexity. Colored vertical lines show group means; black vertical lines mark zero.

References

Bolender, J. (2007). Prehistoric cognition by description: a Russellian approach to the upper Paleolithic. *Biology & Philosophy 22,* 383-399.

Bürkner P (2017). "brms: An R Package for Bayesian Multilevel Models Using Stan." Journal of Statistical Software, 80(1), 1–28. doi:10.18637/jss.v080.i01.

Chen, S., Gil, D., Gaponov, S., Reifegerste, J., Yuditha, T., Tatarinova, T., Progovac, L., & Benítez-Burraco, A. (2024). Linguistic correlates of societal variation: A quantitative analysis. *PloS one, 19*(4), e0300838. https://doi.org/10.1371/journal.pone.0300838

Dryer, Matthew S. & Haspelmath, Martin (eds.) 2013.WALS Online (v2020.4) [Dataset]. Zenodo.https://doi.org/10.5281/zenodo.13950591 (Available online at https://wals.info, Accessed on 2025-11-03.)

Eberhard, David M., Gary F. Simons, and Charles D. Fennig (eds.). 2022. Ethnologue: Languages of the World. Twenty-fifth edition. Dallas, Texas: SIL International. Online version: http://www.ethnologue.com.

Hammarström, Harald & Forkel, Robert & Haspelmath, Martin & Bank, Sebastian. 2022. *Glottolog 4.6.* Leipzig: Max Planck Institute for Evolutionary Anthropology. https://doi.org/10.5281/zenodo.6578297 (Available online at http://glottolog.org, Accessed on 2022-08-25.)

Kirby, K. R., Gray, R. D., Greenhill, S. J., Jordan, F. M., Gomes-Ng, S., Bibiko, H. J., ... & Gavin, M. C. (2016). D-PLACE: A global database of cultural, linguistic and environmental diversity. *PloS one, 1*1(7), e0158391.

Lupyan, G., & Dale, R. (2016). Why Are There Different Languages? The Role of Adaptation in Linguistic Diversity. *Trends in cognitive sciences, 20*(9), 649–660. https://doi.org/10.1016/j.tics.2016.07.005

McWhorter, J. H. (2011). *Linguistic simplicity and complexity: Why do languages undress?* (Vol. 1). Walter de Gruyter.

R Core Team (2021). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing, Vienna, Austria. URL https://www.R-project.org/.

Roberts, S., & Winters, J. (2013). Linguistic diversity and traffic accidents: Lessons from statistical studies of cultural traits. PloS one, 8(8), e70902.

Shcherbakova, O., Michaelis, S. M., Haynie, H. J., Passmore, S., Gast, V., Gray, R. D., Greenhill, S. J., Blasi, D. E., & Skirgård, H. (2023). Societies of strangers do not speak less complex languages. *Science advances, 9*(33), eadf7704. https://doi.org/10.1126/sciadv.adf7704

Sampson, G., Gil, D., and Trudgill, P. (Eds.). (2009). *Language complexity as an evolving variable* (Vol. 13). Oxford University Press.

Skirgård, H., Haynie, H. J., Blasi, D. E., Hammarström, H., Collins, J., Latarche, J. J., ... & Gray, R. D. (2023). Grambank reveals the importance of genealogical constraints on linguistic diversity and highlights the impact of language loss. *Science Advances, 9*(16), eadg6175.

Wray, A. & Grace, G. W. (2007). The consequences of talking to strangers: Evolutionary corollaries of socio-cultural influences on linguistic form. *Lingua 117,* 543-578.